

PRODUCT REQUIREMENT DOCUMENT

Project: Employee Onboarding & Engagement Portal (Web)

1. PRODUCT OVERVIEW

A web-based system to:

1. Automate pre-joining communication
2. Enforce structured onboarding (non-skippable)
3. Ensure policy understanding via assessments
4. Collect employee data with approval workflows
5. Enable AI-driven employee 1:1 conversations
6. Generate HR insights & reports

2. END-TO-END USER FLOW

- **2.1 Pre-Joining Flow**

HR → Adds employee → System triggers:

- T-2 Email (Personal Email)
- Day 0 Emails (Personal + Company Email)

- **2.2 Day 1 Flow (STRICT FLOW)**

1. Login
2. Induction Starts (Auto)
3. Policy 1 → Quiz → Pass Required
4. Policy 2 → Quiz → Pass Required
5. Policy 3 → Quiz → Pass Required
6. Final Assessment (10 Questions)
7. Data Collection Forms
8. Feedback Submission
9. Completion Status

***** User cannot skip or jump steps

3. MODULE-WISE BREAKDOWN

- **3.1 AUTHENTICATION MODULE**

Features:

- Login via User ID + Password
- First login → Force password reset

Acceptance Criteria:

- User cannot access portal without credentials
- Session timeout after inactivity (configurable)

Edge Cases:

- Invalid login → show error
- Forgot password → email reset link

- **3.2 EMAIL AUTOMATION MODULE**

Triggers:

- T-2 days before DOJ
- Day 0 (joining day)

Acceptance Criteria:

- Emails must be sent within ± 5 mins of trigger time
- Email logs must be stored

Edge Cases:

- Email failure → retry 3 times
- Invalid email → notify HR

- **3.3 INDUCTION CONTENT MODULE (HR CONTROLLED)**

Features:

- HR can:
 - ✓ Upload videos/PPT
 - ✓ Create/edit policies
 - ✓ Add MCQs

Acceptance Criteria:

- Changes reflect for new users only (not in-progress users)

Edge Cases:

- Video not loading → fallback message + retry

- **3.4 INDUCTION ENGINE (CORE LOGIC)**

Flow:

FOR EACH POLICY:

1. Play video (auto)
2. Show 3 MCQs
3. Evaluate answers

Rules:

- Minimum 2 correct required

Acceptance Criteria:

- Next policy unlocks ONLY after passing
- Fail → replay same policy

- **3.5 FINAL ASSESSMENT MODULE**

Rules:

- 10 questions
- Passing: ≥70% (configurable)

Acceptance Criteria:

- Fail → restart full induction
- Pass → unlock next module

Edge Cases:

- Browser close → resume from last completed policy

- **3.6 DATA COLLECTION MODULE**

A. ID CARD DETAILS

Fields:

- Name
- Photo
- Blood Group
- Address
- Emergency Contact
- **Flow:**

Employee → HR → Admin

B. COMPLIANCE DETAILS

Fields:

- PF
- ESIC
- Bank Details
- Medical Info
- **Flow:**

Employee → HR → Relevant Team

Acceptance Criteria:

- Cannot proceed without submitting all required fields
- File uploads must be validated

Edge Cases:

- HR rejects → user must edit & resubmit

3.7 APPROVAL WORKFLOW ENGINE

States:

- Pending
- Approved
- Rejected

Acceptance Criteria:

- Notifications triggered on each status change

3.8 AI 1:1 MODULE

- **Trigger:**

HR schedules meeting → portal link shared

- **Flow:**

Screen 1: Select Issue

- Career Growth
- Manager Issues
- Personal
- Policy Confusion
- Work Environment
- Others

Screen 2: Mode Selection

- Chat
- Voice

3.9 AI ENGINE BEHAVIOR

- **Requirements:**

- Conversations tagged to employee (NOT anonymous)
- Adaptive questioning

- **Output:**

- Sentiment score

- Issue category
- Risk level

Acceptance Criteria:

- All conversations stored
- HR dashboard reflects categorized insights

Edge Cases:

- AI failure → fallback to text-based Q&A template

3.10 FEEDBACK MODULE

- **Trigger:**

After onboarding completion

- **Acceptance Criteria:**
- Feedback mandatory before completion

3.11 DASHBOARD & REPORTING

Reports:

- Onboarding completion %
- Assessment scores
- Department-wise insights
- AI sentiment trends

4. SYSTEM RULES (CRITICAL)

- No step skipping allowed
- Sequential flow enforced
- Progress must be saved
- User mapped to department automatically

5. DATA TRACKING

System must capture:

- Login timestamps
- Assessment scores
- Time spent per module
- AI interaction logs
- Approval timelines

6. NON-FUNCTIONAL REQUIREMENTS

- **Performance:**
- <3 sec load time
- **Security:**
- Encrypted storage
- Role-based access
- **Scalability:**
- 1000+ concurrent users

7. TECHNICAL EXPECTATIONS

- **Suggested Architecture:**
- Frontend: React
- Backend: Node.js
- DB: PostgreSQL
- **Services:**
- Auth Service
- Content Service
- Workflow Engine
- AI Service

8. SUCCESS METRICS

- 100% onboarding completion
- Reduced HR manual effort
- Improved employee feedback score
- Early detection of employee issues

9. FUTURE SCOPE

- Mobile app
- HRMS integration (Keka)
- Advanced AI analytics

10. FINAL SUMMARY

This system acts as:

- Onboarding controller
- Policy enforcement engine
- Employee insight system
- HR automation tool